

*Introduction to Quantum Key Distribution*  
*Advanced Certificate in Quantum Computing:*  
*Algorithms and AI/ML*



**Prof. Anshul Jaiswal**

*Department of Electronics and Communication*

*Center for Photonics & Quantum Communication Technology*

*IIT Roorkee*



# Contents

- Introduction
- One-time Pad Encryption
- Ideal Key Generator
- Quantum Key Distribution
- Polarization of Photon
- BB84 Protocol

## Book:

- Ramona Wolf, *Quantum Key Distribution: An Introduction With Exercises*, Springer 2021

# Introduction

- Cryptography:
  - Old concept (*fifth century BC*)
  - Codes and ciphers have evolved throughout history; there have always been attempts *to crack ciphers* and to come up with new and more secure ones.
    - Ciphers: *a secret way of writing*
  - In principle, they can be *cracked*.

Cipher text | P H H W P H D W W K H D S S O H E U H H

- Example of the encryption scheme:
  - *Caesar cipher*
  - The idea is *simple*: every letter of the alphabet is *shifted* by a certain number of letters.
  - Original form → every letter was shifted *three letters*

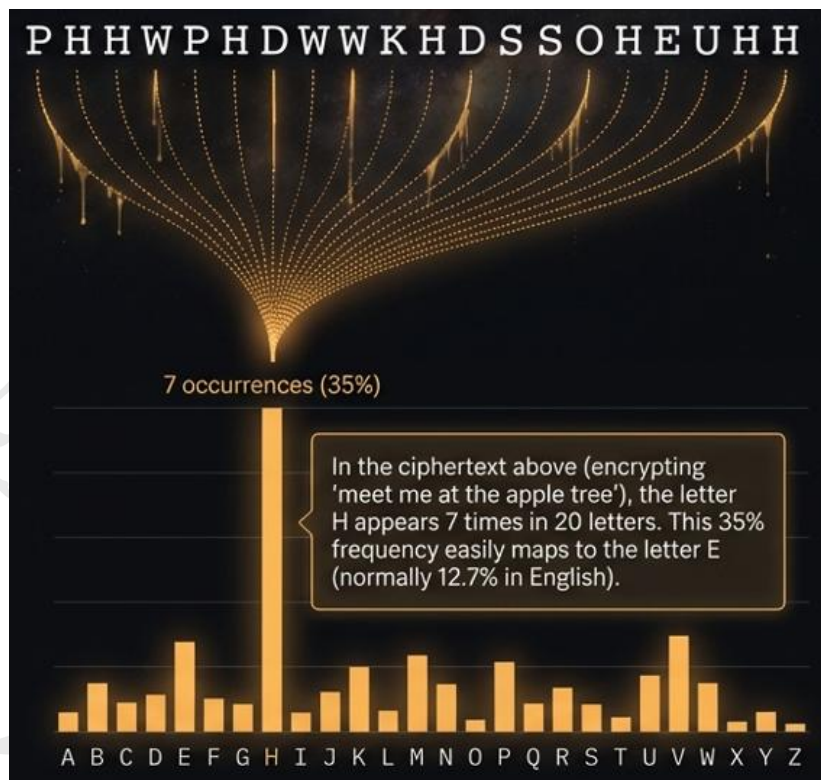
|                 |                                                     |
|-----------------|-----------------------------------------------------|
| Plain alphabet  | a b c d e f g h i j k l m n o p q r s t u v w x y z |
| Cipher alphabet | D E F G H I J K L M N O P Q R S T U V W X Y Z A B C |

# Caesar Cipher

|             |  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------|--|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plain text  |  | m | e | e | t | m | e | a | t | t | h | e | a | p | p | l | e | t | r | e | e |
| Cipher text |  | P | H | H | W | P | H | D | W | W | K | H | D | S | S | O | H | E | U | H | H |

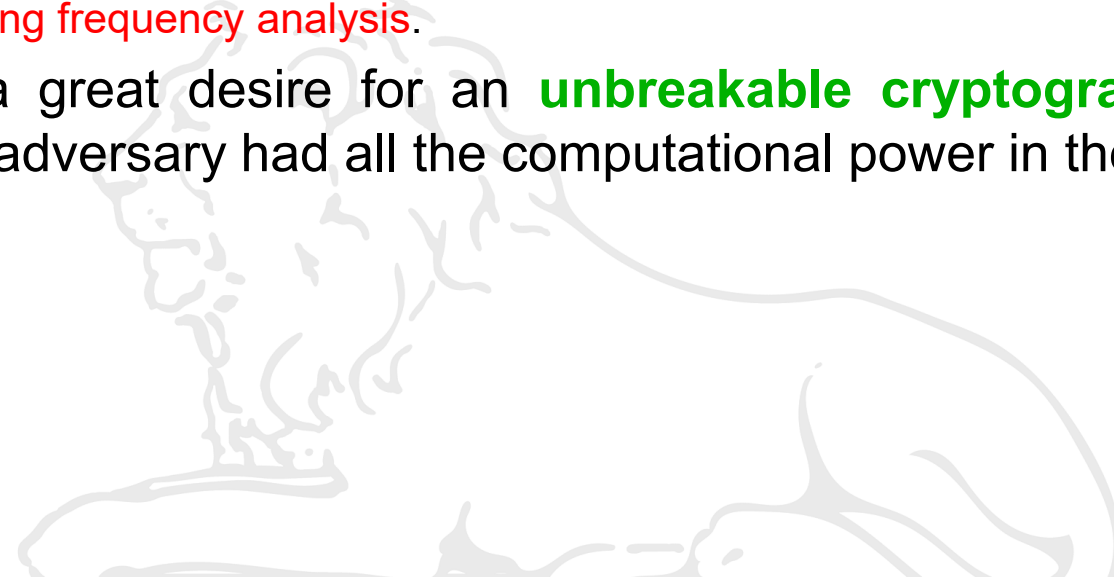
- Can we be **completely sure** that an adversary is unable to get the original message?
  - **Certainly not!**
  - For instance, if he knows about the Caesar cipher, he can **just reverse** the encryption process.
  - But even if he **does not know** about this encryption scheme, there is a way to recover the original message, which works by exploiting the structure of the English language
  - One can **analyze the frequency** with which certain letters appear in a typical English sentence, which is reflected in the frequency of certain letters in the ciphertext.
- In the English language, the letter “E” is the **most frequent one**.

# Caesar Cipher



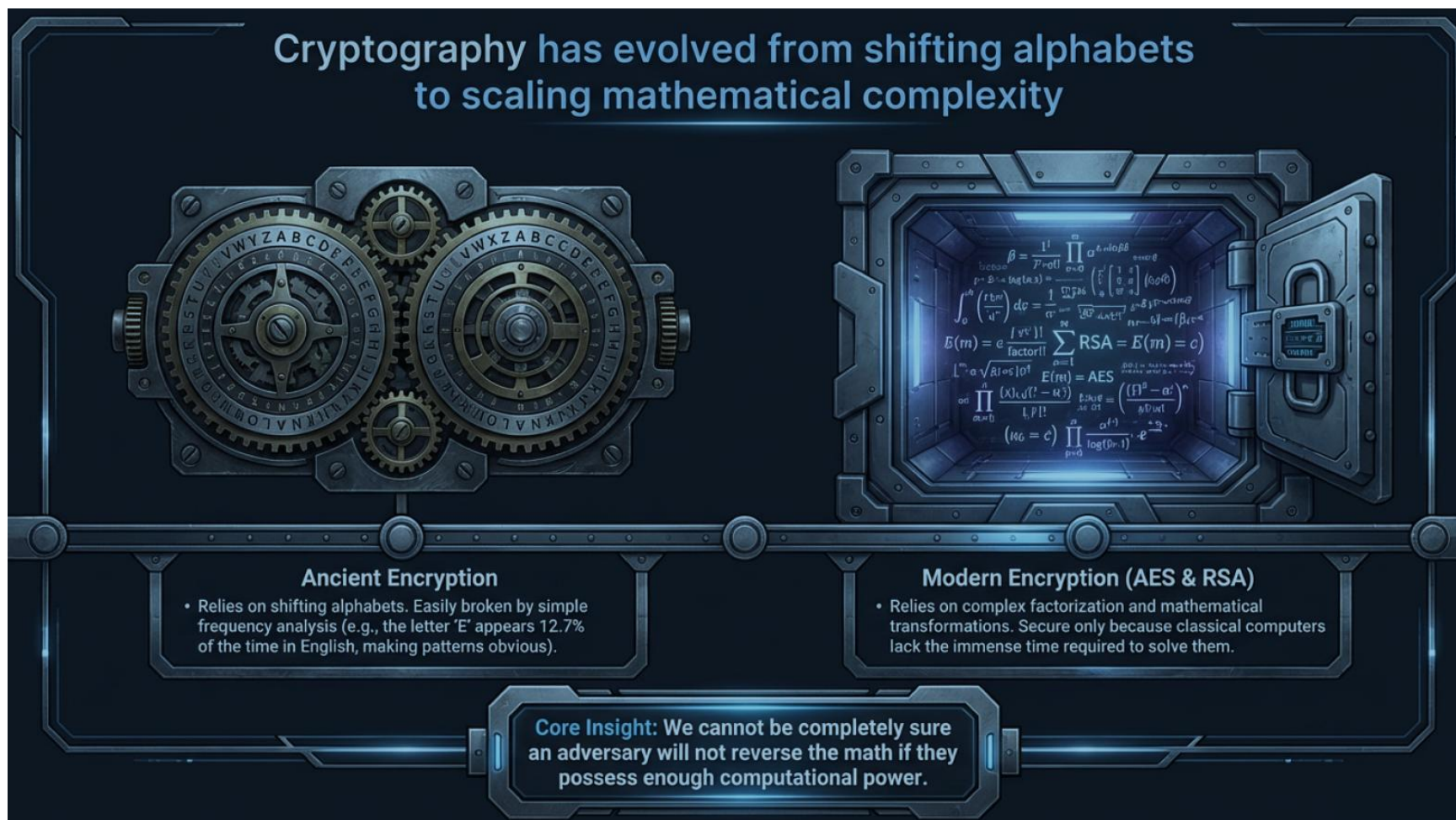
- The message above is a very good example of this:
  - The letter “H” appears seven times in a message of 20 letters, which corresponds to a frequency of 35%.
  - 12.7% for the letter “E” in the English language

# Caesar Cipher

- There are **more complicated encryption schemes** than the Caesar cipher:
    - For instance, one could **shuffle the letters of the alphabet randomly** instead of just shifting them by a certain number.
    - However, this scheme can be cracked in the same way as the Caesar cipher, simply **by applying frequency analysis**.
  - There is a great desire for an **unbreakable cryptographic scheme**, even if an adversary had all the computational power in the universe.
- 
- A faint, light-colored background image of a classical statue, possibly a Greek or Roman figure, is visible behind the text.
- A famous example of how cryptography has had an influence on the outcome of a war is how the British (most famously, Alan Turing) have cracked the Enigma code used by the Germans to communicate during World War II.
    - Please watch: [Click Here](#)



# Present Cipher Techniques



Current cryptography: Security based on **assumptions** about computational hardness

- Although **large-scale quantum computers** are not yet **available**, rapid progress suggests they may emerge within the **next decade or two**.
- Since current cryptographic systems rely on **computational hardness assumptions** that quantum algorithms can break, this creates **a long-term security risk**.

*we are in a “**pre-quantum threat**” phase*

***Any solution?***

- In 1920s, an encryption scheme was proposed which is indeed provably secure:

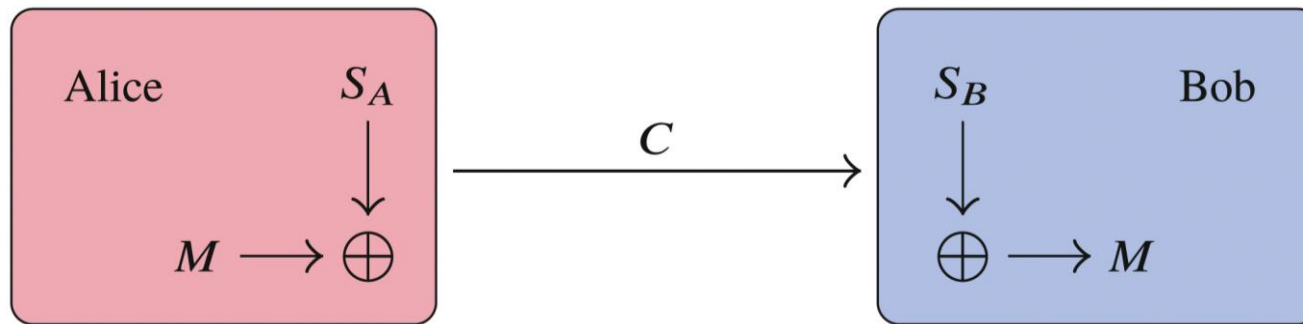
***Vernam cipher, also called one-time pad***



# One-time Pad Encryption

- Preliminaries:
  - Two parties, traditionally called **Alice and Bob**, want to exchange messages over a distance, and they want to be sure that *no one is able to read* these messages.
  - Hypothetical adversary → **Eve**
  - Alice and Bob need **a key to encrypt and decrypt messages** in a way that Eve cannot gain any information about the messages.
  - The word “**key**” now onwards refers to a **string of bits**, i.e., a sequence of 0s and 1s.
  - In the same fashion, whenever we say “**message**”, we mean a message which is in the form of a **bit string**.
- Alice and Bob have successfully created a pair of keys  **$S_A, S_B$** , each of length  $n$ , and want to use it to send a message  **$M = (M_1, M_2, \dots, M_n)$**  of length  $n$  from Alice and Bob.
- $M_1, M_2, \dots, M_n$  represent the individual bits of the message given as a bit string.

# Steps: One-time Pad Encryption



$M$ : Message,  $S_A$ : Key at Alice,  $S_B$ : Key at Bob, and  $C$ : Ciphertext

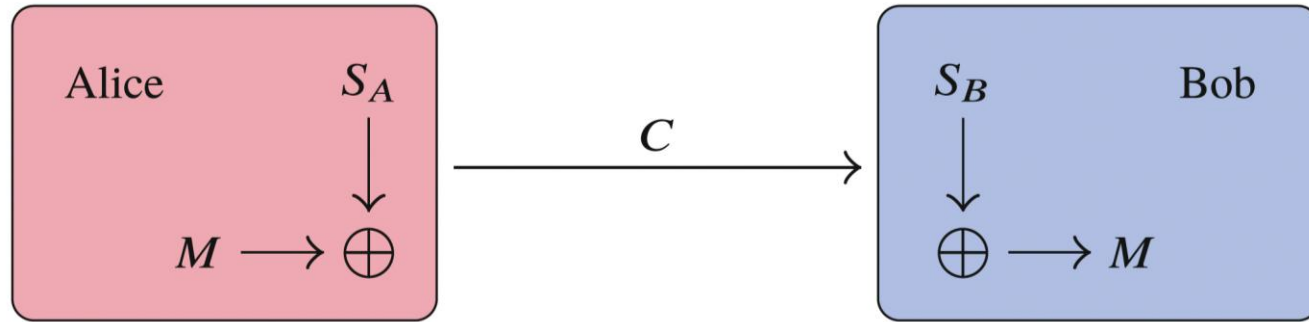
- Alice encrypts the message  $M$  with the key  $S_A$  by doing **binary addition**, which means that each bit of the ciphertext is produced by binary adding a bit from the key to a bit from the message:

$$C_i = M_i \oplus (S_A)_i$$

- The ciphertext  $C = (C_1, C_2, \dots, C_n)$  is sent to Bob. This is done over a public channel, i.e., an adversary has full access to the ciphertext (but **cannot change the message**).
- Bob decodes the message by doing binary addition of the ciphertext and his key  $S_B$ :

$$M_i = C_i \oplus (S_B)_i$$

# Steps: One-time Pad Encryption



- If the **keys are equal**, i.e.,  $S_A = S_B$ , the message Bob holds after decoding is exactly the **same** as the one Alice sent.

$$\begin{aligned}
 M_i &= C_i \oplus (S_B)_i \\
 &= (M_i \oplus (S_A)_i) \oplus (S_B)_i \\
 &= M_i \oplus ((S_A)_i \oplus (S_B)_i) \\
 &= M_i \oplus ((S_A)_i \oplus (S_A)_i) \\
 &= M_i.
 \end{aligned}$$

|         |   |   |   |   |   |   |   |
|---------|---|---|---|---|---|---|---|
| $M$ :   | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| $S_A$ : | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| $C$ :   | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| $S_B$ : | 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| $M'$ :  | 0 | 1 | 1 | 0 | 1 | 0 | 0 |

# Conditions to be met by the Key

- This encryption scheme is indeed information-theoretically secure, which means that the encrypted message (i.e., the ciphertext) **provides no information** about the original message (except the maximum possible **length of the message**), even though the adversary has complete access to the ciphertext.
- However, the key has to **meet certain properties**
  - It has to be **truly random**, which means that the individual key bits are not correlated in any way.
  - It has to be **at least as long as the original message**, which follows directly from the encryption process described above.
  - It can **never be reused in whole or partly**. Otherwise, an adversary can get information about the messages that are sent

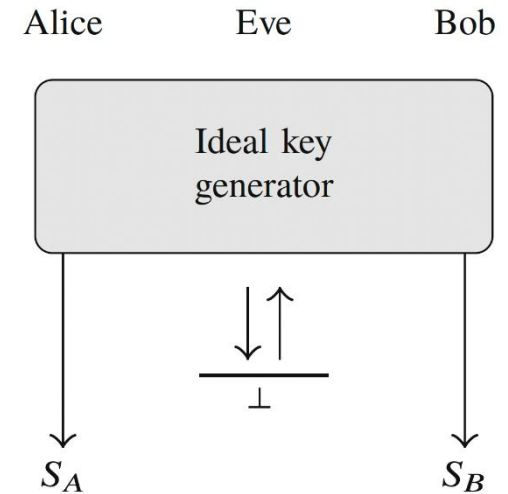
$$\begin{aligned}(C_1 \oplus C_2)_i &= M_i \oplus S_i \oplus N_i \oplus S_i \\ &= M_i \oplus N_i.\end{aligned}$$

- It has to be **kept completely secret**, i.e., the adversary has zero knowledge about the key.

# Ideal Key Generator and Requirements

- Alice and Bob want a device that uses some sort of protocol to **generate keys** for Alice and Bob

- What are the **properties** that we require of such an ideal device?



- The key has to be **correct**, i.e., Alice and Bob should receive a **matching set of keys** to be able to communicate with each other:

$$S_A = S_B \equiv S$$

- The key should be **random**, i.e., every possible bit string of length  $l$  appears with the same probability.

$$\Pr[S = s] = 2^{-l}$$

- Eve **cannot interact** with the device



# Practical Requirements

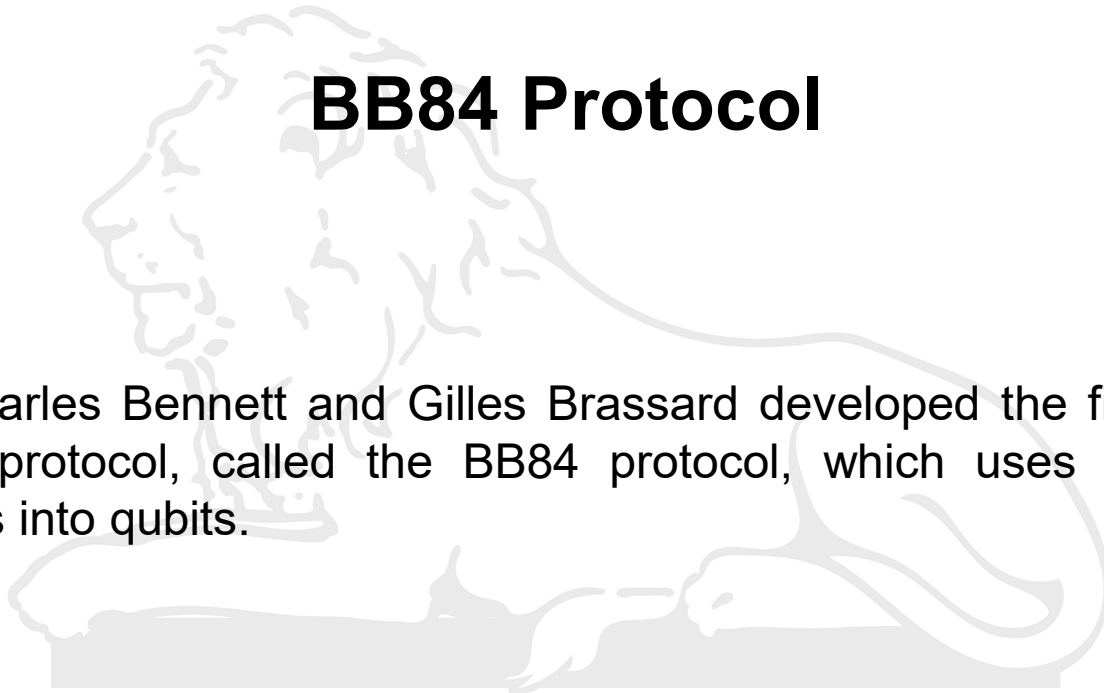
- Unfortunately, we will **never** be able to construct such an ideal key-generating device.
  - For instance, we **cannot completely prevent** Eve from interacting with the box.
- We can weaken the requirements so that the resulting device is not perfect anymore, but **its failure probability is small**.
- The requirements for such **a realistic device** are then changed in the following way:
  - We do not require that the keys the device outputs are always equal, but that the probability for the protocol not to abort and Alice's key differs from Bob's key is very small:
$$\Pr[S_A \neq S_B] \leq \epsilon$$
  - The generated key is close to a perfect key.
    - It is close to a uniformly random key
  - If Eve attack/interact the device, the device will abort the protocol and not output a key.
    - In this way, we lose the property that the device **always** outputs a secure key, but we can be sure that **if a key is produced, this key is secure**.

# Quantum Key Distribution

- The **main goal** of quantum key distribution is to build **a device** where, whenever a key is generated, we can be sure that it is secure according to the previous **(weakened) requirements** so that Alice and Bob never use a corrupted key.

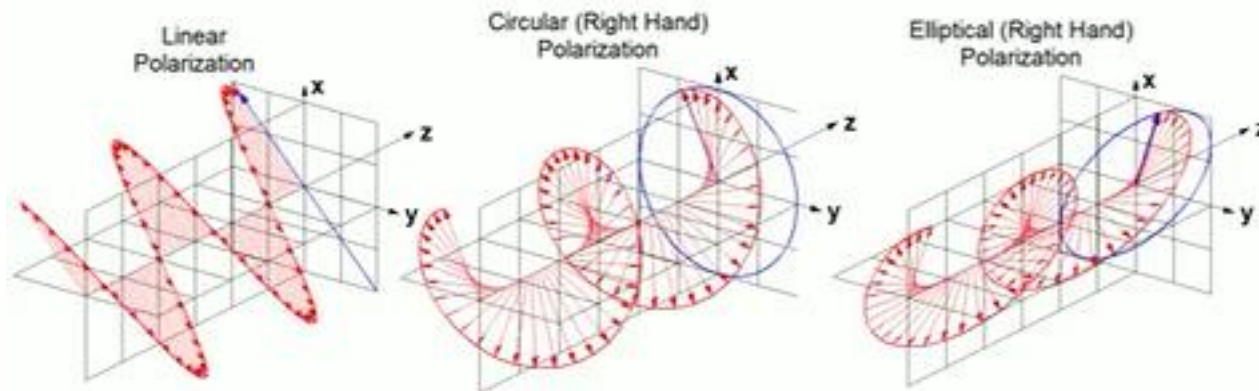
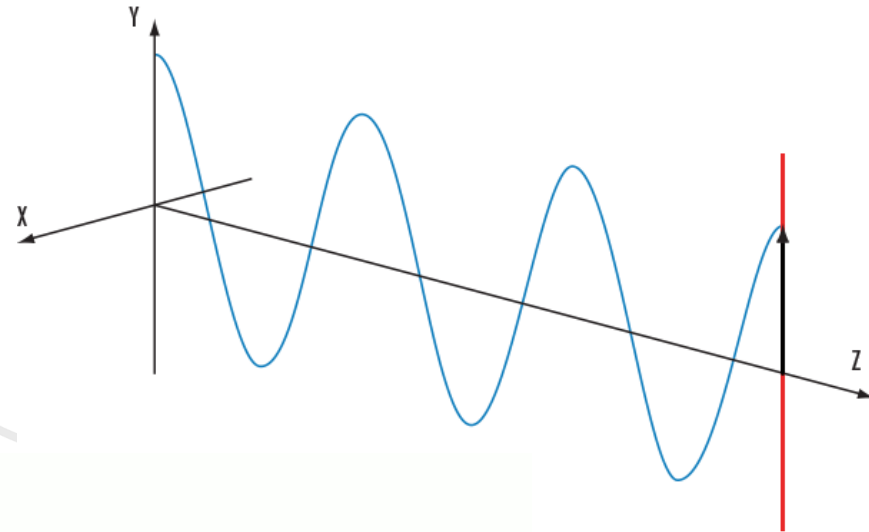
## BB84 Protocol

- In 1984, Charles Bennett and Gilles Brassard developed the first quantum key distribution protocol, called the BB84 protocol, which uses the encoding of classical bits into qubits.



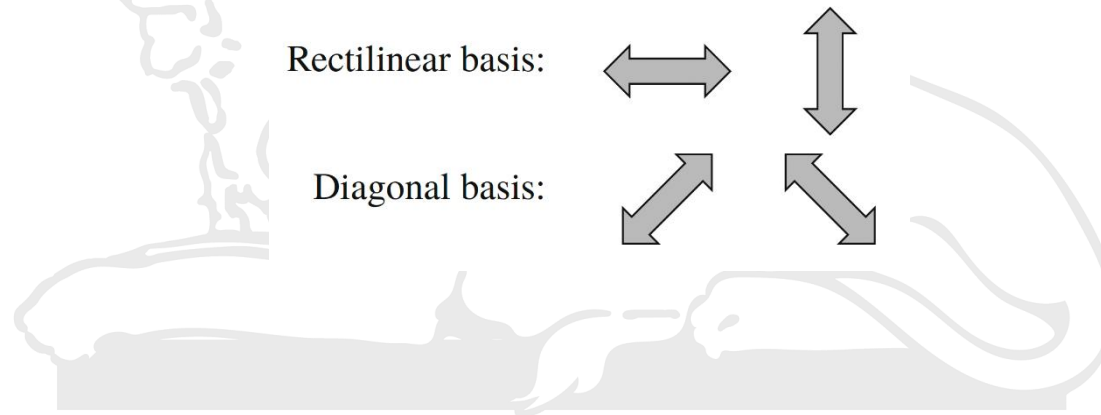
# Polarization of Photons

The polarization of light specifies the geometrical orientation of the oscillation of the electromagnetic field associated with its wave.



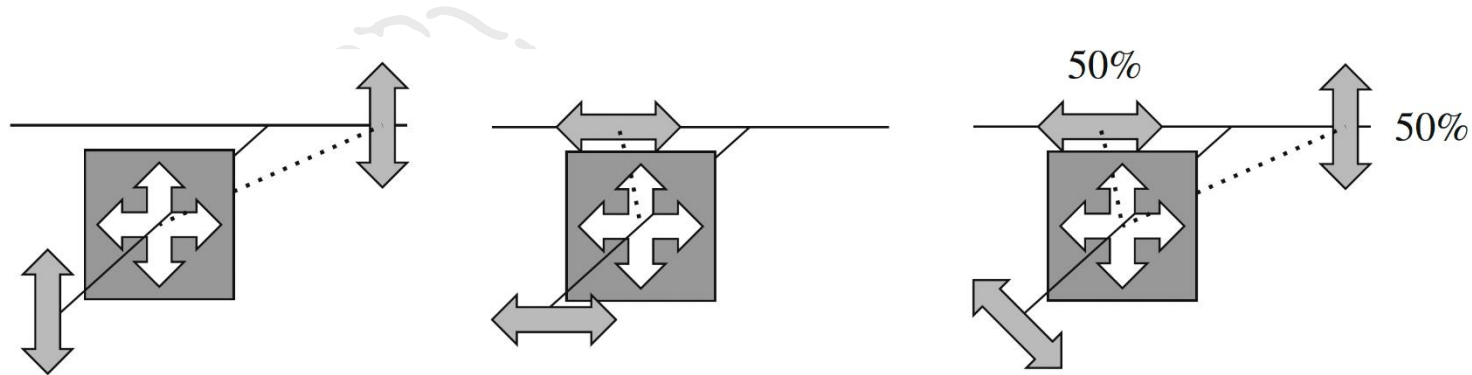
# Polarizations for BB84

- We focus here on **linear polarization**, which means that the field only oscillates in one direction.
- We define two kinds of bases of linear polarization states:
  - The rectilinear basis, which includes **horizontal and vertical** orientations
  - The diagonal basis, which includes orientations **rotated by  $+45^\circ$  and  $-45^\circ$** .



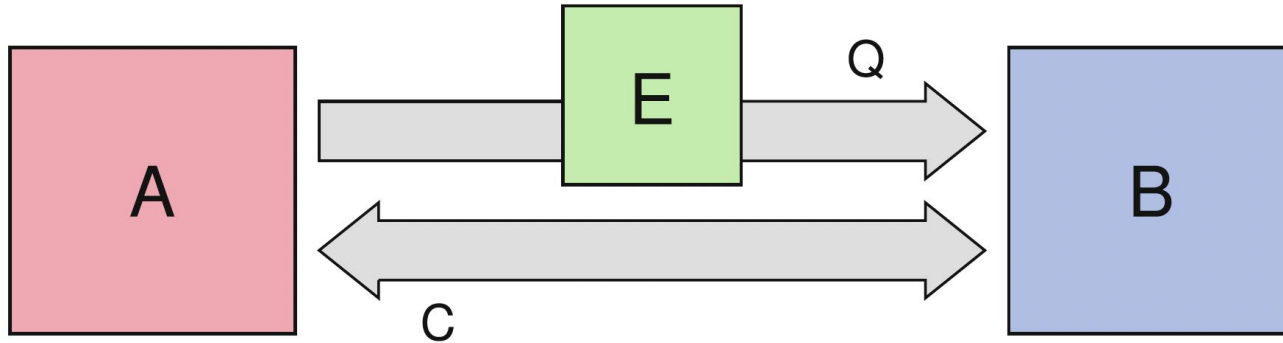
# Polarization Filters

- There exist **filters** to distinguish between horizontally and vertically polarized photons (and, analogously, between  $+45^\circ$  and  $-45^\circ$  rotated photons), as depicted below:



- When a **vertically** (**horizontally**) polarized photon passes through a **rectilinear polarization filter**, it is deflected to the **right** (**left**), as depicted in the left and middle picture.
- When a photon that is polarized in the **diagonal basis** passes through a **rectilinear polarization filter**, it will be deflected to the left **50% of the time** and to the **right 50% of the time**, see the picture on the right

# BB84 Protocol



Basic setting of the BB84 protocol.

- Two parties, Alice (A) and Bob (B), want to establish **a secret key** by using a **quantum channel** (Q) where an adversary, Eve (E), **can tap into without restrictions** and a classical channel (C) where Eve can only listen to the communication but cannot alter it.
- The protocol can be divided into two parts:
  - First, there is a **quantum transmission part**, where Alice and Bob prepare, send, and measure quantum states.
  - The second part is the **classical post-processing part** which is purely classical.



# Quantum Transmission

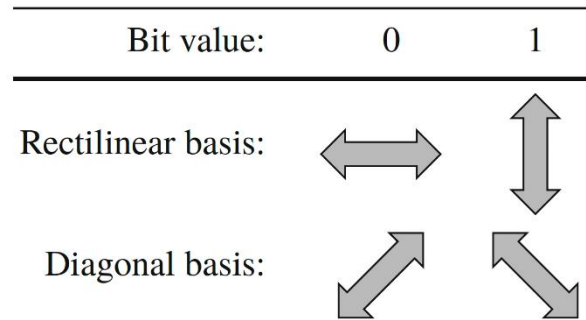
- It includes **encoding and decoding** of classical bits into quantum states and **communication** over a **quantum channel**.
  - Alice chooses a string of  **$N$  random classical** bits  $X_1, \dots, X_N$ .
  - Alice chooses a **random sequence of polarization bases**, where she can either choose the rectilinear basis ( $R$ ) or the diagonal basis ( $D$ ).
  - Alice **encodes** her bit string into a collection of photons with polarization according to the chosen bases, where the encoding works as shown below:

| Bit value:         | 0                                                                                  | 1                                                                                   |
|--------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Rectilinear basis: |  |  |
| Diagonal basis:    |  |  |

- When Bob receives the photons he randomly (**and independently of Alice!**) decides for each photon whether to measure it in the rectilinear or the diagonal basis to obtain classical bits.
- After this step, Alice and Bob both hold a classical bit string, denoted  $X = (X_1, \dots, X_N)$  for Alice and  $Y = (Y_1, \dots, Y_N)$  for Bob.

This is called the **raw key pair**

# Quantum Transmission

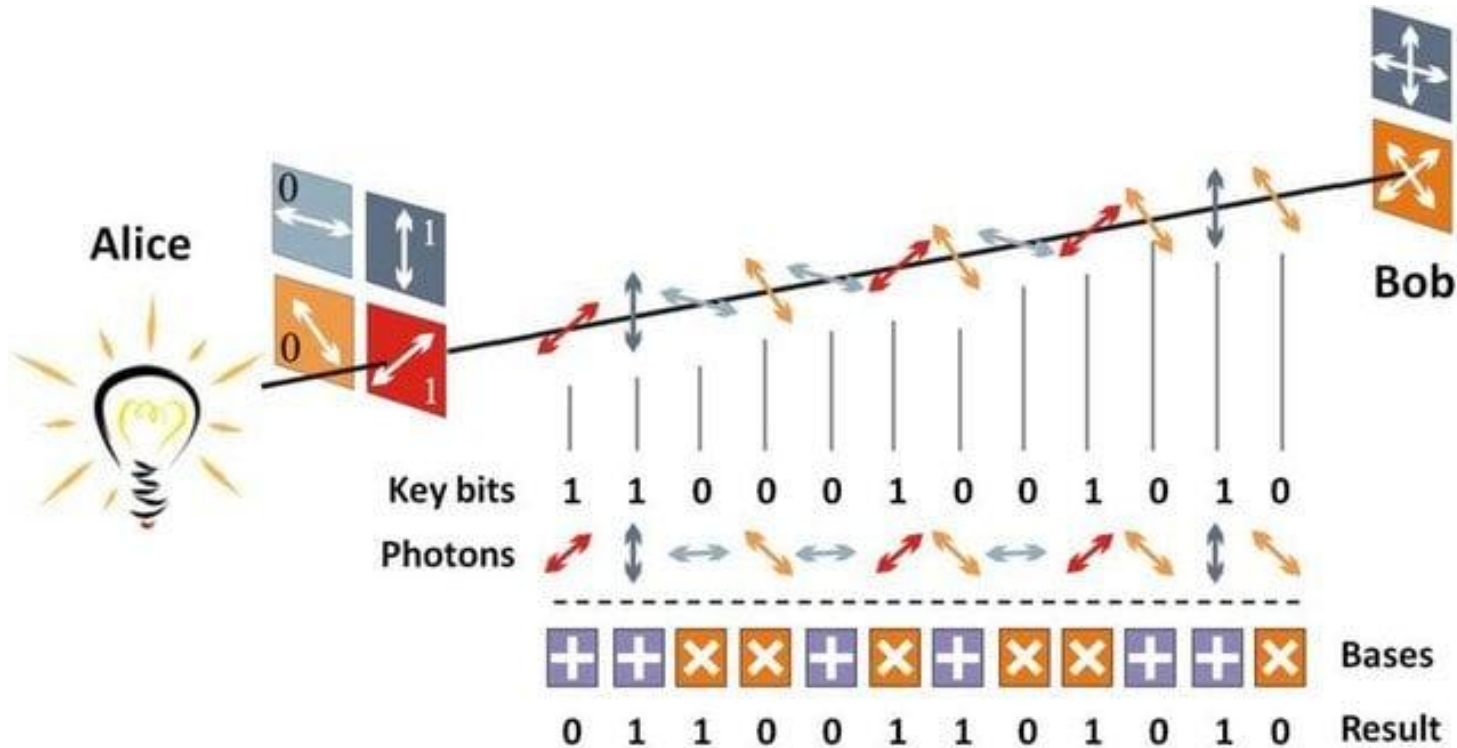


## Quantum transmission

|                         |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alice's random bits     | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| Random sending bases    | D | R | D | D | R | D | R | R | D | D | D | R | D |
| Photons Alice sends     |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Random receiving bases  | D | R | R | D | R | D | D | D | R | D | R | R | D |
| Bits as received by Bob | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |

Raw Keys

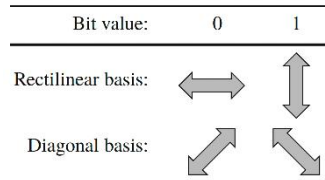
# Quantum Transmission



# Classical Post-Processing

- The rest of the protocol is **purely classical**.
- Alice and Bob exchange **a sequence of classical information** to **transform the bit strings they hold into a shared secret key**.
  - **Sifting Step:**
    - Bob publicly announces the bases he has chosen to measure the photons Alice has sent.
    - Alice compares Bob's bases to the ones she used and says which bases Bob has chosen correctly.
    - Alice and Bob discard all bits for which the encoding and measurement bases are not the same.
  - **Parameter Estimation (PE) Step:**
    - Alice and Bob want to compute a guess for the error rate in the quantum channel
    - Bob reveals some bits of his key at random. In case of no eavesdropping, these bits should be the same as Alice's bits and she confirms them.
    - If the error rate is too high, this indicates that there has been some eavesdropping and Alice and Bob abort the protocol.
    - The bits that have been revealed during this step are discarded afterwards as their information is now public to an eavesdropper.
  - **Information Reconciliation and Privacy Amplification (IRPA) Step:**
    - To further improve the secret key

# Classical Post-Processing



## Quantum transmission

|                         |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Alice's random bits     | 0                                                                                 | 1                                                                                 | 1                                                                                 | 1                                                                                 | 0                                                                                 | 1                                                                                 | 0                                                                                  | 1                                                                                   | 1                                                                                   | 0                                                                                   | 0                                                                                   | 1                                                                                   | 1                                                                                   |
| Random sending bases    | D                                                                                 | R                                                                                 | D                                                                                 | D                                                                                 | R                                                                                 | D                                                                                 | R                                                                                  | R                                                                                   | D                                                                                   | D                                                                                   | D                                                                                   | R                                                                                   | D                                                                                   |
| Photons Alice sends     |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Random receiving bases  | D                                                                                 | R                                                                                 | R                                                                                 | D                                                                                 | R                                                                                 | D                                                                                 | D                                                                                  | D                                                                                   | R                                                                                   | D                                                                                   | R                                                                                   | R                                                                                   | D                                                                                   |
| Bits as received by Bob | 0                                                                                 | 1                                                                                 | 0                                                                                 | 1                                                                                 | 0                                                                                 | 1                                                                                 | 1                                                                                  | 1                                                                                   | 0                                                                                   | 0                                                                                   | 0                                                                                   | 1                                                                                   | 1                                                                                   |

## Classical post-processing

|                                                      |    |    |   |    |    |    |   |   |   |    |   |    |    |
|------------------------------------------------------|----|----|---|----|----|----|---|---|---|----|---|----|----|
| Bob reports bases of received bits                   | D  | R  | R | D  | R  | D  | D | D | R | D  | R | R  | D  |
| Alice says which bases are correct                   | OK | OK |   | OK | OK | OK |   |   |   | OK |   | OK | OK |
| Presumably shared information (without eavesdropper) | 0  | 1  |   | 1  | 0  | 1  |   |   |   | 0  |   | 1  | 1  |
| Bob reveals some key bits                            |    | 1  |   |    |    | 1  |   |   |   |    |   | 1  |    |
| Alice confirms them                                  |    | OK |   |    |    | OK |   |   |   |    |   | OK |    |

Sifted Keys

} PE

## Outcome

|                              |   |  |  |   |   |  |  |  |  |   |  |  |   |
|------------------------------|---|--|--|---|---|--|--|--|--|---|--|--|---|
| Remaining shared secret bits | 0 |  |  | 1 | 0 |  |  |  |  | 0 |  |  | 1 |
|------------------------------|---|--|--|---|---|--|--|--|--|---|--|--|---|

IRPA on these keys